

--1-D ARRAYS USING FUNCTIONS--

1. WAP to find average of array elements.

```
#include<stdio.h>
#include<conio.h>

void input( int [ ], int);
void display(int [ ], int);
float average(int [ ], int);
int main(){

    int a[100], size;
    float avg;
    printf("Enter the size of array:");
    scanf("%d", &size);
    input(a, size);
    printf("\nThe Array Elements are:\n");
    display(a, size);
    avg=average(a, size);
    printf("\nThe Average is:%f\n", avg);

    getch();
    return 0;
}

void input( int a[ ], int n){
    int i;
    for(i=0;i<n; i++){
        printf("Enter a[%d] element:",i);
        scanf("%d", &a[i]);
    }
}

void display( int a[ ], int n){
    int i;
    for(i=0;i<n; i++){
        printf("%d\t",a[i]);
    }
}
```

```
float average(int a[ ], int n){
    int i, Avg, sum=0;
    for(i=0;i<n;i++){
        sum=sum+a[i];
    }
    Avg=(float)sum/n;
    return Avg;
}
```

2. WAP to find Maximum and minimum number in an array.

```
#include<stdio.h>
#include<conio.h>

void input( int [ ], int);
void display(int [ ], int);
int maximum(int [ ], int);
int minimum(int [ ], int);
int main(){
    int a[100], size, max, min;
    printf("Enter the size of array:");

    scanf("%d", &size);
    input(a, size);
    printf("\nThe Array Elements are:\n");
    display(a, size);
    max=maximum(a, size);
    printf("\nThe Maximum element is:%d\n", max);
    min=minimum(a, size);
    printf("\nThe Minimum element is:%d\n", min);
    getch();
    return 0;
}

void input( int a[ ], int n){
    int i;
    for(i=0;i<n; i++){
        printf("Enter a[%d] element:",i);
        scanf("%d", &a[i]);
    }
}
```

```
void display( int a[ ], int n){
    int i;
    for(i=0;i<n; i++){
        printf("%d\t",a[i]);

    }
}

int maximum(int a[ ], int n){
    int i, Max=a[0];

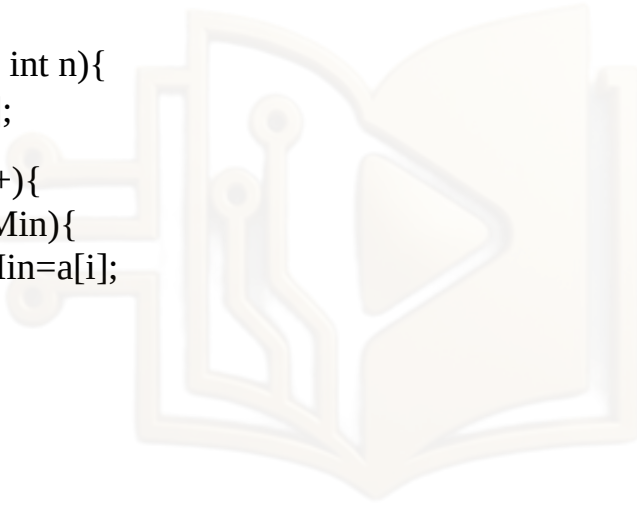
    for(i=0;i<n;i++){
        if(a[i]>Max){
            Max=a[i];
        }
    }

    return Max;
}

int minimum(int a[ ], int n){
    int i, Min=a[0];

    for(i=0;i<n;i++){
        if(a[i]<Min){
            Min=a[i];
        }
    }

    return Min;
}
```



3. WAP to copy one array to another in reverse order.

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
void input( int [ ], int);
```

```
void display(int [ ], int);
```

```
void copy(int [ ], int [ ], int);
```

```
int main(){
```

```
    int a[100], b[100], size;
```

```
    printf("Enter the size of array:");
```

```
    scanf("%d", &size);
    input(a, size);
    printf("\nThe Array Elements are of A are:\n");
    display(a, size);
    copy(a, b, size);
    printf("\nThe Array Elements are of B are:\n");
    display(b, size);
    getch();
    return 0;
}
```

```
void input( int a[ ], int n){
    int i;
    for(i=0;i<n; i++){
        printf("Enter a[%d] element:",i);
        scanf("%d", &a[i]);
    }
}
```

```
void display( int a[ ], int n){
    int i;
    for(i=0;i<n; i++){
        printf("%d\t",a[i]);
    }
}
```

```
void copy(int a[ ], int b[ ], int n){
    int i;
    for(i=n-1; i>=0; i--){
        b[n-i-1]=a[i];
    }
}
```

4. WAP to Add two arrays.

```
#include<stdio.h>
#include<conio.h>
```

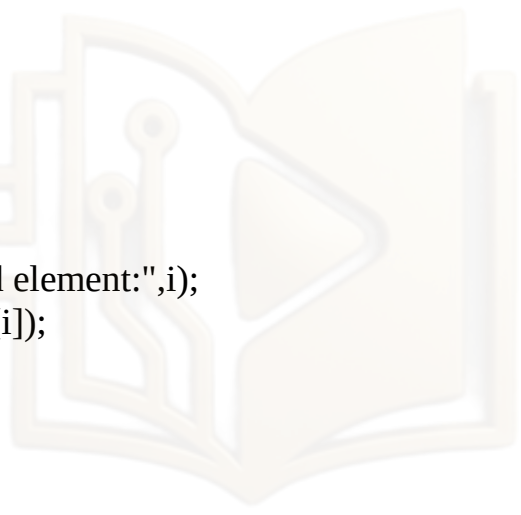
```
void input( int [ ], int);
void display(int [ ], int);
void addition(int[ ], int [ ], int [], int);
```

```
int main(){
    int a[100], b[100], c[100], size;
    printf("Enter the size of array:");
    scanf("%d", &size);
    printf("Enter First array A:\n");
    input(a, size);
    printf("Enter Second array B:\n");
    input(b, size);
    printf("\nThe First Array Elements are:\n");
    display(a, size);
    printf("\nThe Second Array Elements are:\n");
    display(b, size);
    printf("\nThe Added Array Elements are:\n");
    addition(a, b, c, size);
    display(c, size);
    getch();
    return 0;
}

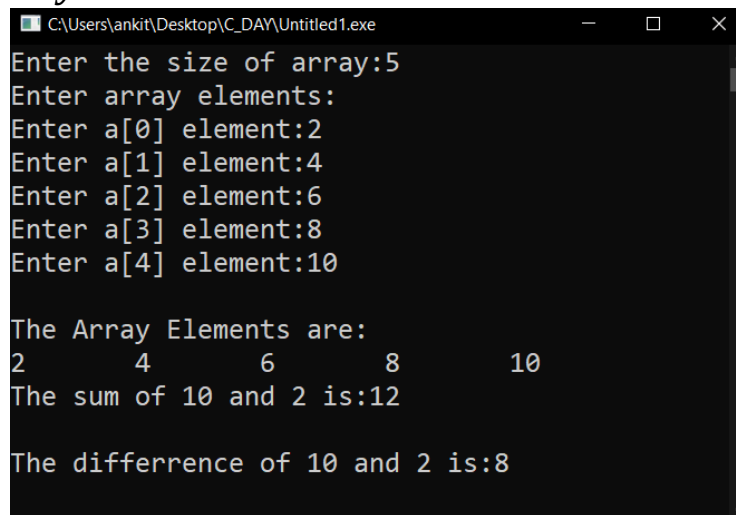
void input( int a[ ], int n){
    int i;
    for(i=0;i<n; i++){
        printf("Enter %d element:",i);
        scanf("%d", &a[i]);
    }
}

void display( int a[ ], int n){
    int i;
    for(i=0;i<n; i++){
        printf("%d\t",a[i]);
    }
}

void addition( int a[ ], int b[ ], int c[ ], int n){
    int i;
    for(i=0;i<n; i++){
        c[i]=a[i] + b[i];
    }
}
```



5. WAP to find sum and difference of maximum and minimum Number in an array.



The screenshot shows a Windows command prompt window titled 'C:\Users\ankit\Desktop\C_DAY\Untitled1.exe'. The user has entered the size of the array as 5 and then entered five elements: 2, 4, 6, 8, and 10. The program then displays the array elements, calculates the sum of the maximum (10) and minimum (2) elements as 12, and calculates the difference as 8.

```
C:\Users\ankit\Desktop\C_DAY\Untitled1.exe
Enter the size of array:5
Enter array elements:
Enter a[0] element:2
Enter a[1] element:4
Enter a[2] element:6
Enter a[3] element:8
Enter a[4] element:10

The Array Elements are:
2      4      6      8      10
The sum of 10 and 2 is:12
The difference of 10 and 2 is:8
```

```
#include<stdio.h>
#include<conio.h>
```

```
void input( int[ ] , int);
void display(int[ ] , int);
void sum_diff(int[ ] , int);
```

```
int main(){
    int a[100], size;
    printf("Enter the size of array:");
    scanf("%d", &size);
    printf("Enter array elements:\n");
    input(a, size);
    printf("\nThe Array Elements are:\n");
    display(a, size);
    sum_diff (a, size);
```

```
    getch();
    return 0;
}
```

```
void input( int a[ ] , int n){
    int i;
    for(i=0;i<n; i++){
        printf("Enter a[%d] element:",i);
        scanf("%d", &a[i]);
    }
}
```

```
void display( int a[ ], int n){
    int i;
    for(i=0;i<n; i++){
        printf("%d\t", a[i]);
    }
}
```

```
void sum_diff(int a[ ], int n){

    int i, max=a[0], min=a[0];

    for(i=0;i<n; i++){
        if(a[i]>max){
            max = a[i];
        }
        else if(a[i] < min){
            min= a[i];
        }
    }
    printf("\nThe sum of %d and %d is:%d\n", max, min, max+min);
    printf("\nThe difference of %d and %d is:%d\n", max, min, max-min);
}
```

6. WAP to find sum of prime numbers in an array.

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
void input( int [ ] , int);
void display(int [ ] , int);
void sum_prime(int[ ] , int);
```

```
int main(){
    int a[100], size;
    printf("Enter the size of array:");
    scanf("%d", &size);
    printf("Enter array elements:\n");
    input(a, size);
    printf("\nThe Array Elements are:\n");
    display(a, size);
    sum_prime (a, size);
}
```

```
    getch();
    return 0;
}

void input( int a[ ], int n){
    int i;
    for(i=0;i<n; i++){
        printf("Enter a[%d] element:",i);
        scanf("%d", &a[i]);
    }
}

void display( int a[ ], int n){
    int i;
    for(i=0;i<n; i++){
        printf("%d\t", a[i]);
    }
}

void sum_prime(int a[ ], int n){
    int i,j, factor, sum=0;
    for(i=0;i<n; i++){
        factor=0;
        for(j=1; j<=a[i]; j++){
            if(a[i]%j==0){
                factor++;
            }
        }
        if(factor==2){
            sum=sum + a[i];
        }
    }
    printf("\nThe Sum of Prime Number is %d ", sum);
}
```

